SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Dry Run Evaluation

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 2 – Dry Run Evaluation
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	Batch 1	**Date:**	10-10-2026

Code of Conduct

I Prabhakaran, 1912260199 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully; answer all four questions (25 marks each, 100 in total).
- Show every dry-run step: the initial queue, stack, priority queue or belief state, each iteration, your assumptions and a clear final answer, and finish with the interpretation asked for.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Dry Run Evaluation rubric: understanding of the search problem, state expansion and dry run execution, application of the search algorithm, accuracy of the solution path, and presentation and logical reasoning.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end). Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q2**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**
9. Uninformed Search Strategies	Dry runs of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality

Problem understanding: The vacuum world has two squares A and B. The agent starts at A, both squares are dirty, dirt does not reappear, and the performance horizon is T = 6. Each clean square at the end of a step gives +1, while every Left/Right move costs p (p = 1 unless otherwise stated).

(a) PEAS Description

Component	Description
Performance measure	+1 for every clean square at the end of each time step; −p for every Left/Right move over T = 6 steps.
Environment	Two-square vacuum world, A and B; initially both dirty; dirt never reappears.
Actuators	Suck, Left, Right, NoOp.
Sensors	Perceives its current location and whether the current square is Dirty or Clean.

(b) Agent X – Simple Reflex Trace

Method: Apply Agent X's rule at each step: if the current square is Dirty, Suck; otherwise move to the other square.

Step	Location / Percept	Action	Clean squares at end	Move penalty	Step score
1	A, Dirty	Suck	A	0	1
2	A, Clean	Right	A	1	0
3	B, Dirty	Suck	A, B	0	2
4	B, Clean	Left	A, B	1	1
5	A, Clean	Right	A, B	1	1
6	B, Clean	Left	A, B	1	1
Total				4	5

Total score for p = 1: 1 + 0 + 2 + 1 + 1 + 1 = 5.

(c) Agent Y – Reflex with Internal State Trace

Method: Agent Y cleans a dirty square; after remembering that both squares have been cleaned, it chooses NoOp instead of moving.

Step	Location / Percept	Action	Clean squares at end	Move penalty	Step score
1	A, Dirty	Suck	A	0	1
2	A, Clean; not both cleaned yet	Right	A	1	0
3	B, Dirty	Suck	A, B	0	2
4	B, Clean; remembers both clean	NoOp	A, B	0	2
5	B, Clean; remembers both clean	NoOp	A, B	0	2
6	B, Clean; remembers both clean	NoOp	A, B	0	2
Total				1	9

Total score for p = 1: 1 + 0 + 2 + 2 + 2 + 2 = 9.

(d) Total Scores as Functions of p

Agent X makes four moves (steps 2, 4, 5 and 6). Therefore:

$S_X(p) = 1 + (1 - p) + 2 + (1 - p) + (2 - p) + (2 - p) = 9 - 4p.$

Agent Y makes one move (step 2). Therefore:

$S_Y(p) = 1 + (1 - p) + 2 + 2 + 2 + 2 = 10 - p$.

Y is strictly better than X when $10 - p > 9 - 4p$, so $3p > -1$ and $p > -1/3$. For the stated meaningful range $p \geq 0$, Agent Y is strictly better for every non-negative move penalty.

(e) Rationality Interpretation

Agent Y is rational under this performance measure because, after both squares are clean, continuing to move gives no cleaning benefit and creates an unnecessary penalty. Its internal state lets it remember that the whole environment has already been cleaned and select NoOp. Agent X cannot match Y because X uses only the current percept; when it sees a clean square, it always moves, even when the other square has already been cleaned. Thus the percept sequence and stored state affect the rational action choice.

Question 2: Task Environment Analysis and Agent Selection

Method: Classify each environment on the six given properties, assign 1 to the harder value and 0 to the easier value (semi-dynamic = 0.5), then sum the scores.

(a) Classification of the Six Properties

Property	E1: Autonomous Taxi	E2: Conveyor Robot	E3: Chess Agent
Observability	Partially observable – traffic and road conditions are not fully known.	Partially observable – camera view may not reveal every relevant defect/state.	Fully observable – board position and legal moves are visible.
Number of agents	Multi-agent – other road users affect the taxi.	Multi-agent – other agents/people may affect the workspace.	Multi-agent – an opponent actively chooses actions.
Determinism	Stochastic – traffic and events are uncertain.	Stochastic – moving parts/defects and timing are uncertain.	Deterministic – moves have known game rules, though the opponent is strategic.
Episodic / Sequential	Sequential – current decisions affect later driving states.	Sequential – picking actions occur continuously over a moving belt.	Sequential – each move changes the future game state.
Static / Dynamic	Dynamic – traffic and surroundings change while acting.	Dynamic – the conveyor keeps moving during operation.	Semi-dynamic – board state is static between moves, but the clock changes.
Discrete / Continuous	Continuous – position, speed and time vary continuously.	Continuous – position/timing on the moving belt vary continuously.	Discrete – finite board states and moves.

(b) Difficulty Scores

Using the question's convention: partially observable = 1, multi-agent = 1, stochastic = 1, sequential = 1, dynamic = 1, semi-dynamic = 0.5, continuous = 1; easier values score 0.

Environment	Observability	Agents	Determinism	Sequential	Static/Dynamic	Discrete/Continuous	Total
E1 Taxi	1	1	1	1	1	1	6
E2 Conveyor robot	1	1	1	1	1	1	6
E3 Chess	0	1	0	1	0.5	0	2.5

Ranking: E1 and E2 are tied as hardest (6 points each), followed by E3 as easiest (2.5 points).

(c) Agent Architecture

Hardest environment (E1/E2): A learning agent is most suitable because these environments are partially observable, stochastic, sequential, dynamic and continuous. Learning can improve decisions from experience and adapt to changing conditions. For the taxi specifically, a utility-based component is also useful for balancing safety, travel time and other performance factors.

Easiest environment (E3): A goal-based agent is suitable for chess because the goal is to reach a winning game state, and the environment is fully observable and discrete. The agent can search possible moves toward the goal while considering the opponent's actions.

(d) Interpretation

Agent complexity is determined by environmental properties rather than the domain name alone. Partial observability requires the agent to maintain a model or belief about hidden information; stochasticity requires handling uncertainty; sequential and dynamic settings require planning and continual updating; continuous settings require control over changing values. Therefore, harder properties generally require richer internal state, planning, utility evaluation or learning.

Question 3: Turing Test Data Across the History of AI

Method: Correct-identification rate = correctly identified as machine / 50 × 100. Fooling rate = (50 − correctly identified) / 50 × 100.

(a) Correct-Identification and Fooling Rates

System	Correctly identified	Correct-identification rate	Fooling rate
ELIZA	47/50	94%	6%
Chatbot B	36/50	72%	28%
System C	29/50	58%	42%

Interpretation: System C has the highest fooling rate (42%), meaning judges most often mistook it for a human among the three systems.

(b) Turing 70% Criterion and Distance from 50% Chance

The reference says an average interrogator would have no more than a 70% chance of correct identification. Therefore, a system is consistent with the criterion when its correct-identification rate is ≤ 70%.

System	Correct-identification rate	Meets ≤70% criterion?	Distance from 50% chance level
ELIZA	94%	No	44 percentage points above 50%
Chatbot B	72%	No	22 percentage points above 50%
System C	58%	Yes	8 percentage points above 50%

(c) Improvement in Fooling Rate

ELIZA → System C: fooling rate changes from 6% to 42%. Improvement = 42 − 6 = 36 percentage points. Ratio = 42/6 = 7, so System C's fooling rate is 7 times ELIZA's.

Chatbot B → System C: fooling rate changes from 28% to 42%. Improvement = 42 − 28 = 14 percentage points.

(d) Interpretation

The Turing Test measures acting humanly: it evaluates whether a machine's conversational behavior can be mistaken for human behavior. A high fooling rate does not by itself prove that System C is thinking or acting rationally. It only shows that its observed conversational behavior can resemble human responses. This separates the acting-humanly approach from claims about genuine understanding, internal reasoning or rational action.

Question 4: Rational Route Choice under Uncertainty

Given: deadline = 55 minutes and U = 100 − T − 20L, where L = 1 if arrival is after 55 minutes and L = 0 otherwise.

Method: First calculate utility for each possible outcome. Then use expected utility: EU = Σ P(outcome) × U(outcome). Choose the route with the highest expected utility.

(a) Utility of Each Route

Route	Outcome	Time T	Late L	Utility U = 100 − T − 20L
R1 Highway	Normal	30	0	70
R1 Highway	Disrupted	70	1	10 (late)
R2 City road	Normal	45	0	55
R2 City road	Disrupted	60	1	20 (late)
R3 Ring road	Normal	50	0	50
R3 Ring road	Disrupted	50	0	50 (no disruption)

(b) Expected Utility and Rational Choice

R1: EU = 0.70(70) + 0.30(10) = 49 + 3 = 52.

R2: EU = 0.90(55) + 0.10(20) = 49.5 + 2 = 51.5.

R3: EU = 1.00(50) = 50.

Therefore, R1 (Highway) is the rational choice because it has the highest expected utility: 52.

(c) New Percept: Highway Disruption Probability = 0.80

Updated R1 expected utility = 0.20(70) + 0.80(10) = 14 + 8 = 22.

R2 remains 51.5 and R3 remains 50. Therefore, after the new radio alert, the rational choice changes from R1 to R2 (City road), because 51.5 is now the highest expected utility.

(d) Actual Outcome and Rationality

The route chosen in part (b) was R1. If the disruption occurs, T = 70 and the taxi is late, so L = 1.

Actual utility = 100 − 70 − 20(1) = 10.

The agent was not irrational. Rationality is judged using the information available when the decision was made and the expected performance measure, not by demanding perfect knowledge of future events. An omniscient agent would know in advance that the disruption would occur and could choose differently, but an ordinary rational agent cannot be expected to know an uncertain future event.

(e) Interpretation

Rationality depends on the percept sequence because the agent should use all relevant information available at the time of action. It also depends on the performance measure because the best action is the one expected to maximize the stated utility. Part (c) changes the decision because the radio alert changes the agent's percept and therefore changes the probability distribution used in expected-utility calculation, even though the physical roads themselves have not changed.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Accuracy of Solution Path	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Q. No. / Criteria	Understanding of Search Problem	State Expansion and Dry Run Execution	Application of Search Algorithm	Accuracy of Solution Path	Accuracy of Solution Path	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____